LESSON 01 / 06

The problem: no trig function is one-to-one

A function has an inverse only if it is one-to-one (injective). Every trig function fails globally: $\sin 0 = \sin \pi = 0$, $\tan 0 = \tan \pi = 0$, and so on. Graphically, a horizontal line $y = c$ with $|c| < 1$ crosses $y = \sin x$ infinitely many times. So "$\sin^{-1}$" cannot exist on all of $\mathbb{R}$ until we ***cut the domain down to a strip on which sine is one-to-one and takes every value in*** $[-1, 1]$.

INTUITION

Inverse = undo: the one question that decides everything

$y = \sin^{-1} x$ means exactly: "y is the angle (in the chosen strip) whose sine is x". So $\sin(\sin^{-1} x) = x$ is *automatic* on the domain of $\sin^{-1}$ — but $\sin^{-1}(\sin y) = y$ only for y inside the chosen strip. ***All the difficulty of this chapter lives on that one asymmetry.*** The strip is a convention — but once fixed (by the official principal-value ranges below), every formula is determined, and every "simplify" question is a branch-check question.

The six principal values (memorise the ranges — they are the whole syllabus)

Function	Domain	Principal range (the branch)
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1} x$	$\lvert x\rvert \geq 1$	$[0, \pi] \setminus \{\pi/2\}$
$\csc^{-1} x$	$\lvert x\rvert \geq 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$

Reading the table correctly: $\sin^{-1}$ is the *inverse of the restriction* $\sin : [-\pi/2, \pi/2] \to [-1, 1]$; $\cos^{-1}$ *inverts* $\cos : [0, \pi] \to [-1, 1]$; $\tan^{-1}$ *inverts* $\tan : (-\pi/2, \pi/2) \to \mathbb{R}$. *The half-open endpoints for* $\tan^{-1}$ *reflect the vertical asymptotes of* $\tan$.

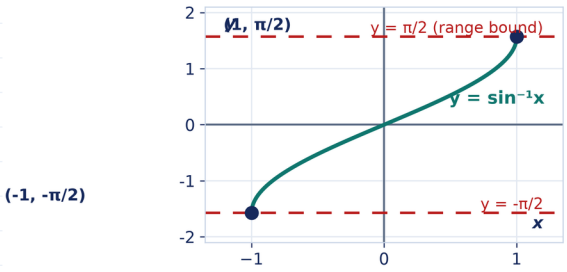
ASSUMPTIONS & CONVENTIONS

Symbols and conventions used everywhere below

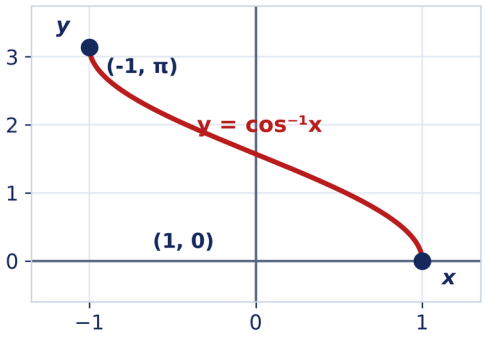
- $\sin^{-1} x$ means the ***inverse function*** (arcsine), never $\csc x = (\sin x)^{-1}$. Write $\sin^{-1}$ / arcsin for the inverse; never write $\sin^{-1} x$ for $1/\sin x$.
- All inputs x are real; ranges above are in ***radians***. Degree-mode answers convert at the boundary only.
- "$\sin^{-1}(\sin y) = y$"-type claims always silently assume y lies in the principal range of the outer inverse. Part of the answer is *stating* that assumption.
- $\cot^{-1}$ takes range $(0, \pi)$ (the JEE convention); some calculus texts use $(-\pi/2, 0) \cup (0, \pi/2]$ — the value of $\cot^{-1} x + \tan^{-1} x$ then changes. We fix $(0, \pi)$ throughout and note the caveat here once.

Graphs: domain on the x-axis, branch on the y-axis

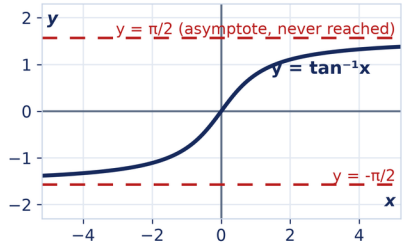
y = sin⁻¹x : domain [-1, 1] ⊂ x-axis, branch [-π/2, π/2] ⊂ y-axis



y = cos⁻¹x : range [0, π] — note cos⁻¹1 = 0, cos⁻¹(-1) = π



y = tan⁻¹x : range (-π/2, π/2) is OPEN — the asymptotes are not values



Three graph-reading habits: (i) the **branch bounds are horizontal lines on the y-axis** — they constrain the output; (ii) endpoints may be attained ($\sin^{-1}$, $\cos^{-1}$) or not ($\tan^{-1}$); (iii) $\sin^{-1}$ and $\tan^{-1}$ are **odd** functions ($\sin^{-1}(-x) = -\sin^{-1}x$, $\tan^{-1}(-x) = -\tan^{-1}x$), visible as origin-symmetry; $\cos^{-1}$ is **neither** — its graph is not symmetric.

The three complementary-pair identities (state + prove)

KEY RESULT

Complementary sums — valid on the stated domains

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \qquad |x| \le 1$$

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, \qquad x \in \mathbb{R}$$

$$\sec^{-1}x + \csc^{-1}x = \frac{\pi}{2}, \qquad |x| \ge 1$$

SHORT DERIVATION

Proof of the first — the template for all three

Let $\theta = \sin^{-1}x$, so $\theta \in [-\pi/2, \pi/2]$ and $\sin\theta = x$. Then $\cos(\pi/2 - \theta) = \sin\theta = x$, and $\pi/2 - \theta \in [0, \pi]$ — which is exactly the principal range of $\cos^{-1}$. Hence $\cos^{-1}x = \pi/2 - \theta = \pi/2 - \sin^{-1}x$. ∎

The second: $\cot(\pi/2 - \theta) = \tan\theta$, and the range bookkeeping ($\pi/2 - \tan^{-1}x \in (0, \pi)$) is identical. Same for the third. **The mechanism is always: co-function identity + range verification.** The range check is not decoration — it is what makes the inverse principal.

Standard values (the x-values worth instant recall)

x	$\sin^{-1}x$	x	$\cos^{-1}x$	x	$\tan^{-1}x$
0	0	0	$\pi/2$	0	0
1/2	$\pi/6$	1/2	$\pi/3$	$1/\sqrt{3}$	$\pi/6$
$1/\sqrt{2}$	$\pi/4$	$1/\sqrt{2}$	$\pi/4$	1	$\pi/4$
$\sqrt{3}/2$	$\pi/3$	$\sqrt{3}/2$	$\pi/6$	$\sqrt{3}$	$\pi/3$
1	$\pi/2$	1	0	—	$\to \pi/2$
$-1/2$	$-\pi/6$	$-1/2$	$2\pi/3$	-1	$-\pi/4$
-1	$-\pi/2$	-1	π	—	$\to -\pi/2$

Note the systematic pattern: $\sin^{-1}$ and $\cos^{-1}$ of the same x always sum to $\pi/2$; $\tan^{-1}$ of the reciprocal of $\sqrt{3}$-values is the complementary angle. Negative inputs: $\sin^{-1}$ and $\tan^{-1}$ give negative outputs (odd functions); $\cos^{-1}(-x) = \pi - \cos^{-1}x$ (output stays in $[0, \pi]$).

COMMON TRAP

Common traps — foundations

- $\sin^{-1}(\sin\theta) = \theta$ is **false** for e.g. $\theta = \frac{3\pi}{4}$; the left side is $\pi/4$. Trap of the entire chapter.
- $\sin(\sin^{-1}x) = x$ holds for all $x \in [-1, 1]$; $\sin^{-1}(\sin x) = x$ holds only for $x \in [-\pi/2, \pi/2]$.
- Domain before value: $\sin^{-1}2$ is **undefined** in $\mathbb{R}$, not "$\pi/2$". $\cos^{-1}(1.01)$ does not exist.
- $\tan^{-1}x$ never equals $\pm\pi/2$; writing $\tan^{-1}x = \pi/2$ "as $x \to \infty$" is a limit, not a value.
- $(\sin x)^{-1} = \csc x \neq \sin^{-1}x$. The notation is genuinely ambiguous in print; our convention is stated in the ASSUMPTIONS box and used without exception.
- $\cos^{-1}x$ is not an odd function: $\cos^{-1}(-1/2) = 2\pi/3 \neq -\pi/3$.

LESSON 02 / 06

The one skill: reduce the angle into the principal branch

For every "inverse of a trig" the question is: *which angle in the principal range has the same trig value as x?* The answer is never "x itself" unless x already lies in the branch. The reliable method is a **two-step reduction** (shown for $\sin^{-1}(\sin x)$; the others are identical in spirit):

KEY RESULT

Reduction algorithm for $f^{-1}(fx)$ — reduce, then read the branch

1. **Reduce mod the period** of f: replace x by $t = x - 2k\pi$ (sine/cosine) or $t = x - k\pi$ (tangent/cotangent), with $k \in \mathbb{Z}$ chosen so that t falls in one full period.
2. **Read the branch**: find the unique angle y in the principal range with $f(y) = f(t)$; that y is the answer.

$\sin^{-1}(\sin x)$ in closed piecewise form

Reduce x to $t \in [-\pi/2, 3\pi/2)$ by subtracting $2\pi k$. Then the closed piecewise read is just two cases:

$$\sin^{-1}(\sin t) = t \quad \text{for } t \in [-\pi/2, \pi/2]$$

$$\sin^{-1}(\sin t) = \pi - t \quad \text{for } t \in (\pi/2, 3\pi/2)$$

Why $\pi - t$: for $t \in (\pi/2, 3\pi/2)$, the angle $\pi - t \in (-\pi/2, \pi/2)$ and $\sin(\pi - t) = \sin t$ — it is the mirror angle on the branch, produced by the symmetry of the sine wave about $t = \pi/2$.

WORKED EXAMPLE

Worked — $\sin^{-1}\left(\sin\frac{7\pi}{6}\right)$ and $\sin^{-1}\left(\sin\frac{5\pi}{4}\right)$

$\frac{7\pi}{6} = \pi + \frac{\pi}{6}$ is in $(\pi/2, 3\pi/2)$: answer $= \pi - \frac{7\pi}{6} = -\frac{\pi}{6}$. Check: $\sin\frac{7\pi}{6} = -\frac{1}{2} = \sin(-\frac{\pi}{6})$ ✓, and $-\frac{\pi}{6}$ is on the branch ✓.

$\frac{5\pi}{4} \in (\pi/2, 3\pi/2)$: answer $= \pi - \frac{5\pi}{4} = -\frac{\pi}{4}$. Check: $\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2} = \sin(-\frac{\pi}{4})$ ✓.

For $\sin^{-1}(\sin\frac{9\pi}{5})$: first reduce — $\frac{9\pi}{5} - 2\pi = -\frac{\pi}{5} \in [-\pi/2, \pi/2]$, so the answer is $-\frac{\pi}{5}$. **Reduce first, read second** — skipping the reduction is how wrong signs appear.

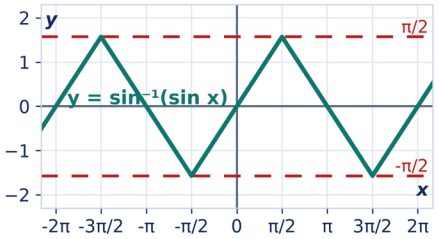
$\cos^{-1}(\cos x)$

Reduce to $t \in [0, 2\pi)$; then $\cos^{-1}(\cos t) = t$ for $t \in [0, \pi]$ and $= 2\pi - t$ for $t \in (\pi, 2\pi)$ (mirror about $t = 0$ on the circle: $2\pi - t \in (0, \pi)$ has the same cosine). E.g. $\cos^{-1}(\cos\frac{7\pi}{4}) = 2\pi - \frac{7\pi}{4} = \frac{\pi}{4}$.

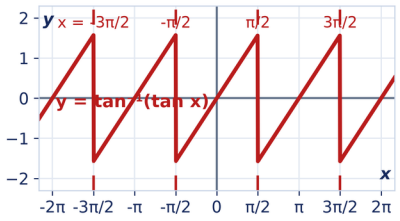
$\tan^{-1}(\tan x)$

Reduce to $t \in (-\pi/2, \pi/2)$ by subtracting πk (period π, not 2π!): $\tan^{-1}(\tan x) = x - \pi k$ with the unique $k \in \mathbb{Z}$ landing the result in $(-\pi/2, \pi/2)$. E.g. $\tan^{-1}(\tan\frac{3\pi}{4}) = \frac{3\pi}{4} - \pi = -\frac{\pi}{4}$; $\tan^{-1}(\tan 5) = 5 - 2\pi$ (since $5 - 2\pi \approx -1.28 \in (-\pi/2, \pi/2)$). Same for $\cot^{-1}(\cot x)$ with output forced into $(0, \pi)$.

$\sin^{-1}(\sin x)$: a sawtooth oscillating INSIDE the branch [-π/2, π/2]



$\tan^{-1}(\tan x)$: period π, jumps at odd multiples of π/2 (undefined there)



WORKED EXAMPLE

Worked — $\sin^{-1}\left(\sin\frac{7\pi}{6}\right)$ and $\sin^{-1}\left(\sin\frac{5\pi}{4}\right)$

$\frac{7\pi}{6} = \pi + \frac{\pi}{6}$ is in $(\pi/2, 3\pi/2)$: answer $= \pi - \frac{7\pi}{6} = -\frac{\pi}{6}$. Check: $\sin\frac{7\pi}{6} = -\frac{1}{2} = \sin(-\frac{\pi}{6})$ ✓, and $-\frac{\pi}{6}$ is on the branch ✓.

$\frac{5\pi}{4} \in (\pi/2, 3\pi/2)$: answer $= \pi - \frac{5\pi}{4} = -\frac{\pi}{4}$. Check: $\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2} = \sin(-\frac{\pi}{4})$ ✓.

For $\sin^{-1}(\sin\frac{9\pi}{5})$: first reduce — $\frac{9\pi}{5} - 2\pi = -\frac{\pi}{5} \in [-\pi/2, \pi/2]$, so the answer is $-\frac{\pi}{5}$. **Reduce first, read second** — skipping the reduction is how wrong signs appear.

The reverse compositions: trig of an inverse

$\sin(\tan^{-1}x)$, $\tan(\cos^{-1}x)$ and friends never need formulas — draw the **reference right triangle**:

STANDARD APPLICATION

The triangle method — every trig-of-inverse in one picture

Given $\theta = \tan^{-1}x$: θ is the acute-or-negative angle with $\tan\theta = x = \frac{x}{1}$. Draw a right triangle with opposite $= x$, adjacent $= 1$, hypotenuse $= \sqrt{1+x^2}$ (the sign of x fixes the quadrant; the triangle gives magnitudes). Read off:

$$\sin(\tan^{-1}x) = \frac{x}{\sqrt{1+x^2}}, \qquad \cos(\tan^{-1}x) = \frac{1}{\sqrt{1+x^2}}$$

valid for all real x (both sides odd/even as they must be). Likewise with $\theta = \sin^{-1}x$: opposite $= x$, hypotenuse $= 1$, adjacent $= \sqrt{1-x^2}$ (non-negative because $\theta \in [-\pi/2, \pi/2]$, where $\cos\theta \geq 0$):

$$\cos(\sin^{-1}x) = \sqrt{1-x^2},$$

$$\tan(\sin^{-1}x) = \frac{x}{\sqrt{1-x^2}}, \quad |x| < 1$$

For $\theta = \cos^{-1}x$ ($\theta \in [0, \pi]$): adjacent $= x$, hypotenuse $= 1$, opposite $= \sqrt{1-x^2}$ (again $\sin\theta \geq 0$ on the branch):

$$\sin(\cos^{-1}x) = \sqrt{1-x^2},$$

$$\tan(\cos^{-1}x) = \frac{\sqrt{1-x^2}}{x}, \quad x \neq 0$$

The **sign of the square root** is decided by the branch, not by taste: $\sqrt{1-x^2}$ appears with a $+$ because both $\cos(\sin^{-1}x)$ and $\sin(\cos^{-1}x)$ are non-negative on their branches. If you ever need $\cos(\sin^{-1}x)$ for an angle outside a branch — you don't, that's the point of branches.

SHORT DERIVATION

Worked — simplify $\tan(\sin^{-1}x)$ for $x \in (-1, 1)$ and evaluate $\sin\left(\cos^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13}\right)$

First part: from the triangle, $\frac{x}{\sqrt{1-x^2}}$.

Second: let $A = \cos^{-1}\frac{3}{5} \in [0, \pi]$, $B = \sin^{-1}\frac{5}{13} \in [-\pi/2, \pi/2]$. Then $\sin A = \frac{4}{5}$ (positive on the branch!), $\cos B = \frac{12}{13}$ (positive on the branch!). So

$$\sin(A+B)$$

$$= \sin A\cos B + \cos A\sin B$$

$$= \frac{4}{5}\cdot\frac{12}{13} + \frac{3}{5}\cdot\frac{5}{13}$$

$$= \frac{48+15}{65} = \frac{63}{65}.$$

The trap: $\sin A = \pm 4/5$ in general; the branch of $\cos^{-1}$ forces $+$.

COMMON TRAP

Common traps — branch reading

- $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \frac{\pi}{4}$, **not** $\frac{3\pi}{4}$ — $\frac{3\pi}{4}$ is outside $[-\pi/2, \pi/2]$.
- $\tan^{-1}(\tan x)$ has period π and is undefined at $x = \frac{\pi}{2} + k\pi$; $\sin^{-1}(\sin x)$ has period 2π.
- $\cos(\sin^{-1}x) = +\sqrt{1-x^2}$ always (branch $\Rightarrow$ cosine ≥ 0); never $\pm$.
- $\tan(\cos^{-1}x)$ is positive for $x > 0$ and negative for $x < 0$ — the triangle gives magnitudes, the branch gives signs.
- Reducing *before* reading: $\sin^{-1}\left(\sin\frac{9\pi}{5}\right) = -\frac{\pi}{5}$ requires subtracting 2π first; treating $\frac{9\pi}{5}$ as "in $(\pi/2, 3\pi/2)$" is wrong because the two-case read assumes the reduction already happened.

LESSON 03 / 06

The master formula and why conditions are unavoidable

Take tangents of $\tan^{-1}x \pm \tan^{-1}y$ using the compound-angle formula. The tangent of the answer is never the issue — the issue is which **branch of** $\tan^{-1}$ the sum lands in. Since each of $\tan^{-1}x, \tan^{-1}y \in (-\pi/2, \pi/2)$, their sum lies in $(-\pi, \pi)$; and $\tan^{-1}$ returns values only in $(-\pi/2, \pi/2)$. Whenever the true sum exits that window, a $\pm\pi$ correction is mandatory.

KEY RESULT

Addition and subtraction — statement with corrections

For all real x, y:

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\frac{x+y}{1-xy} \quad \text{if } xy < 1$$

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\frac{x+y}{1-xy}$$

$$\text{if } xy > 1 \text{ and } x > 0$$

$$\tan^{-1}x + \tan^{-1}y = -\pi + \tan^{-1}\frac{x+y}{1-xy}$$

$$\text{if } xy > 1 \text{ and } x < 0$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy} \quad \text{if } xy > -1$$

with the analogous $\pm\pi$ corrections when $xy < -1$ (sign chosen so the final answer lies in $(-\pi, \pi)$, matching the true sum).

Domain wall: the fraction is undefined at $xy = 1$ (sum $= \pm\pi/2$) and $xy = -1$ (difference $= \pm\pi/2$) — those are boundary cases, handled by direct branch arithmetic, not by the formula.

SHORT DERIVATION

Derivation of the corrections — sign of the cosine decides

Let $A = \tan^{-1}x$, $B = \tan^{-1}y$, $S = A + B \in (-\pi, \pi)$. First, $\tan S = \frac{x+y}{1-xy}$ (compound angle), so $S = \tan^{-1}\frac{x+y}{1-xy} + k\pi$ for some integer $k \in \{-1, 0, 1\}$. Which k? Ask where S sits on $(-\pi, \pi)$ via its cosine:

$$\cos S = \cos A \cos B\,(1 - \tan A \tan B)$$

$$\cos A \cos B\,(1 - xy)$$

and $\cos A, \cos B > 0$ strictly (both branches inside $(-\pi/2, \pi/2)$), so $\operatorname{sign}(\cos S) = \operatorname{sign}(1 - xy)$:

- $xy < 1$: $\cos S > 0$ forces $S \in (-\pi/2, \pi/2)$ — the branch itself: $k = 0$.
- $xy > 1$, $x > 0$ (then $y > 0$): $S \in (\pi/2, \pi)$: $k = +1$.
- $xy > 1$, $x < 0$ (then $y < 0$): $S \in (-\pi, -\pi/2)$: $k = -1$.
- $xy = 1$: $\cos S = 0$ and $S = \pm\pi/2$ — compute directly, the fraction is void.

■ (Every line of the table is machine-checked at adversarial points in `math_checks.py`.)

WORKED EXAMPLE

Worked — $\tan^{-1}2+\tan^{-1}3$ and the sign trap $x=2, y=3$; $xy=6>1$ and $x>0$, so the correction fires:

$$\tan^{-1}2+\tan^{-1}3=\pi+\tan^{-1}\tfrac{5}{1-6}=\pi+\tan^{-1}(-1)$$

$$\pi-\tfrac{\pi}{4}=\tfrac{3\pi}{4}.$$

Without the correction you would answer $-\pi/4$ — wrong quadrant. Sanity: $\tan^{-1}2\approx1.107$, $\tan^{-1}3\approx1.249$, sum $\approx2.356=\frac{3\pi}{4}$ ✓.

By contrast $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}$: $xy=\frac{1}{6}<1$, no correction:

$$\tan^{-1}\tfrac{1}{2}+\tan^{-1}\tfrac{1}{3}$$

$$=\tan^{-1}\left(\tfrac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{6}}\right)$$

$$=\tan^{-1}1=\tfrac{\pi}{4}.$$

The $2\tan^{-1}x$ chain — three faces, three conditions

Set $y=x$ in the addition formula, and separately take sin/cos of $2\tan^{-1}x$ via the triangle of sheet 02. All three forms are JEE staples:

KEY RESULT

$2\tan^{-1}x$ in three forms (conditions are part of the formula)

$$2\tan^{-1}x=\tan^{-1}\tfrac{2x}{1-x^2}\quad |x|<1$$

$$2\tan^{-1}x=\pi+\tan^{-1}\tfrac{2x}{1-x^2}\quad x>1;$$

$$=-\pi+\tan^{-1}\tfrac{2x}{1-x^2}\quad x<-1$$

$$2\tan^{-1}x=\sin^{-1}\tfrac{2x}{1+x^2}\quad |x|\le1$$

$$2\tan^{-1}x=\pi-\sin^{-1}\tfrac{2x}{1+x^2}\quad x>1;$$

$$=-\pi-\sin^{-1}\tfrac{2x}{1+x^2}\quad x<-1$$

$$2\tan^{-1}x=\cos^{-1}\tfrac{1-x^2}{1+x^2}\quad x\ge0;$$

$$=-\cos^{-1}\tfrac{1-x^2}{1+x^2}\quad x<0$$

SHORT DERIVATION

Where the $\sin^{-1}$ face comes from

Let $\theta=\tan^{-1}x$, so $\tan\theta=x$. Then $\sin2\theta=\frac{2\tan\theta}{1+\tan^2\theta}=\frac{2x}{1+x^2}$. Now branch-check: for $|x|\le1$, $2\theta\in[-\pi/2,\pi/2]$ and $\sin^{-1}(\sin2\theta)=2\theta$ ✓. For $x>1$, $2\theta\in(\pi/2,\pi)$, so $2\theta=\pi-\sin^{-1}(\sin2\theta)$ ✓ (mirror-angle read of sheet 02). The $\cos^{-1}$ face: $\cos2\theta=\frac{1-x^2}{1+x^2}$, and $2\theta\in[0,\pi]$ exactly when $\theta\ge0$, i.e. $x\ge0$ ✓. Same computation, three branch bookkeepings.

STANDARD APPLICATION

Standard applications — simplification and telescoping

1. **Simplify a sum of two arctans**: check xy against 1 first, then apply the table. E.g. $\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}$; $xy=\frac{1}{56}<1\to$ $\tan^{-1}\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{56}}=\tan^{-1}\frac{15/56}{55/56}=\tan^{-1}\frac{3}{11}$.

1. **Sums that telescope**: recognise $\tan^{-1}\frac{a-b}{1+ab}=\tan^{-1}a-\tan^{-1}b$ (when $ab>-1$). E.g.

$$\sum_{k=1}^{n}\tan^{-1}\tfrac{1}{1+k+k^2}=\sum_{k=1}^{n}[\tan^{-1}(k+1)-\tan^{-1}k]$$

$$\tan^{-1}(n+1)-\tfrac{\pi}{4}.$$

because $1+k+k^2=1+k(k+1)$ and $\frac{(k+1)-k}{1+k(k+1)}=\frac{1}{1+k+k^2}$ ✓. (Olympiad sheets push this hard; the exam-visible core is the identity itself.)

1. **Equations**: $2\tan^{-1}x=\sin^{-1}x$-style problems: convert both sides to one inverse via the chain, then use **injectivity of the outer inverse**, never of $x\mapsto$ expression.

COMMON TRAP

Common traps — the addition formulas

- Forgetting the $\pm\pi$: $\tan^{-1}2+\tan^{-1}3=\frac{3\pi}{4}$, not $-\frac{\pi}{4}$. Rule of thumb: two positive arctans of numbers >1 each... more precisely if $x,y>0$ and $xy>1$ the sum exceeds $\frac{\pi}{2}$ and needs $+\pi$.
- Writing $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$ with **no condition at all** — the identity is false at e.g. $x=y=2$ as written.
- $xy=1$ and $xy=-1$ are boundary cases: the fraction $\frac{x+y}{1-xy}$ explodes while the true sum is $\pm\frac{\pi}{2}$ or $\pm\frac{\pi}{2}\mp\pi$ — compute from the branch directly: if $x=y=1$, $\tan^{-1}1+\tan^{-1}1=\frac{\pi}{2}$ ✓.
- $2\tan^{-1}x=\sin^{-1}\frac{2x}{1+x^2}$ fails for $x>1$: e.g. $x=2$: LHS $=2\tan^{-1}2\approx 2.214$, but $\sin^{-1}\frac{4}{5}\approx 0.927$ — the $\pi-$ face is the true one ($\pi-0.927\approx 2.214$ ✓).
- In telescoping, check $ab>-1$ before using $\tan^{-1}\frac{a-b}{1+ab}=\tan^{-1}a-\tan^{-1}b$; for positive a,b it always holds.

LESSON 04 / 06

$\sin^{-1}x \pm \sin^{-1}y$ — derived, not recited

Let $A=\sin^{-1}x$, $B=\sin^{-1}y$, so $A,B\in[-\pi/2,\pi/2]$ and $A+B\in[-\pi,\pi]$. Take the sine of the sum:

$$\sin(A+B)=x\sqrt{1-y^2}+y\sqrt{1-x^2}=:\sigma$$

(the square roots are non-negative precisely because of the branches). Then $A+B=\sin^{-1}\sigma$ **if** $A+B\in[-\pi/2,\pi/2]$, else the mirror-angle correction applies. The branch test reduces to the sign of

$$\cos(A+B)=\sqrt{1-x^2}\sqrt{1-y^2}-xy$$

which is ≥ 0 exactly when $xy\leq 0$ or $x^2+y^2\leq 1$.

KEY RESULT

The sum formulas, with the branch conditions

$$\sin^{-1}x+\sin^{-1}y=\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

$$\text{if } xy\leq 0 \text{ or } x^2+y^2\leq 1$$

$$\sin^{-1}x+\sin^{-1}y=\pi-\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

$$\text{if } x>0,\ y>0,\ x^2+y^2>1$$

$$\sin^{-1}x+\sin^{-1}y$$

$$-\pi-\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$$

$$\text{if } x<0,\ y<0,\ x^2+y^2>1$$

Difference:
$\sin^{-1}x-\sin^{-1}y=\sin^{-1}x+\sin^{-1}(-y)$ — run the same table with $y\mapsto -y$. All statements assume $|x|,|y|\leq 1$.

WORKED EXAMPLE

Worked — $\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{8}{17}$: which case?

$x=\frac{3}{5}$, $y=\frac{8}{17}$: $xy>0$ and $x^2+y^2=\frac{9}{25}+\frac{64}{289}\approx 0.36+0.22=0.58<1$ → **first case**, no correction. Compute $\sigma=\frac{3}{5}\cdot\frac{15}{17}+\frac{8}{17}\cdot\frac{4}{5}=\frac{45}{85}+\frac{32}{85}=\frac{77}{85}$. So the sum equals $\sin^{-1}\frac{77}{85}$.

Cross-check numerically: $\sin^{-1}0.6\approx 0.6435$, $\sin^{-1}(8/17)\approx 0.4899$, sum ≈ 1.1334; $\sin^{-1}(77/85)\approx 1.1334$ ✓.

Now the other regime: $\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{12}{13}$: $x^2+y^2=0.64+0.852>1$, both positive → second case. $\sigma=\frac{4}{5}\cdot\frac{5}{13}+\frac{12}{13}\cdot\frac{3}{5}=\frac{20+36}{65}=\frac{56}{65}$. So the sum equals $\pi-\sin^{-1}\frac{56}{65}$. Numerically the two sides agree:
$\sin^{-1}0.8+\sin^{-1}(12/13)\approx 0.9273+1.176$(
, and
$\pi-\sin^{-1}(56/65)\approx 3.1416-1.0383=2.1$(
✓. Without the correction you would quote $\sin^{-1}\frac{56}{65}\approx 1.0383$ — not the sum.

cos⁻¹x ± cos⁻¹y

Same machinery with $A, B \in [0, \pi]$: $\cos(A \pm B) = xy \mp \sqrt{1-x^2}\sqrt{1-y^2}$, and the sum $A + B \in [0, 2\pi]$ needs one π-type correction when $A + B > \pi$ (i.e. when $x + y < 0$). The clean exam form:

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$$

$$\text{if } x + y \geq 0$$

$$\cos^{-1} x + \cos^{-1} y$$

$$2\pi - \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$$

$$\text{if } x + y < 0$$

$$\cos^{-1} x - \cos^{-1} y = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$$

$$\text{if } x \leq y$$

and $= -\cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$ if $x > y$ (the difference is negative then). Also worth memorising as a one-liner: $\cos^{-1} x - \sin^{-1} y = \cos^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x}\right.$ for $x, y \geq 0$... the safest habit: **derive the needed formula on the spot from** $\cos(A \pm B)$ **or** $\sin(A \pm B)$ **plus a branch check**, and use the table only as a time-saver you have already verified once.

ASSUMPTIONS & CONVENTIONS

Assumptions packed into the four formulas above

$|x|, |y| \leq 1$ throughout; the un-specified square roots $\sqrt{1-x^2}$, $\sqrt{1-y^2}$ always mean the **non-negative** root; all angles in radians; outputs of $\sin^{-1}$ in $[-\pi/2, \pi/2]$ and of $\cos^{-1}$ in $[0, \pi]$ without exception. When a problem statement gives angles outside these windows, the problem is a sheet-02 reduction problem first.

The simplification toolkit — patterns that actually appear

STANDARD APPLICATION

Five standard simplification patterns

1. **Sum of two arctans** → sheet 03 table (check xy vs 1). Classic: $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \frac{\pi}{4} + \frac{3\pi}{4} = \pi$.
2. **Pair a sine-inverse with a cosine-inverse of the same argument**: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ collapses the pair instantly — scan for it before computing anything.
3. **Square-root absorption**: $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ for $x > 0$ recognises the half-angle: with $x = \tan\theta$ this is $\tan^{-1}\frac{1-\cos\theta}{\sin\theta} = \frac{\theta}{2} = \frac{1}{2}\tan^{-1} x$. Setting $x = \tan\theta$ (or $x = \sin\theta$, $x = \cos\theta$) to expose a compound angle is the standard move.
4. **Convert to one inverse**: $\sin^{-1} x = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$, $\cos^{-1} x = \tan^{-1}\frac{\sqrt{1-x^2}}{x}$ (branch-checked), then use the arctan table.
5. **Monotone reading for equations**: each inverse is strictly monotone on its domain ($\sin^{-1}$, $\tan^{-1}$ increasing; $\cos^{-1}$ decreasing), so $f^{-1}(\text{stuff}) = c$ may be replaced by "$\text{stuff} = f(c)$" — injectivity of the outer function is what licenses the step.

COMMON TRAP

Common traps — addition formulas

- $\sin^{-1}x+\sin^{-1}y=\sin^{-1}(x\sqrt{1-y^2}+y\sqrt{1-x^2})$ is **conditional**. Test the regime first: at $x=y=\frac{3}{5}$ one has $x^2+y^2=0.72<1$ so the plain case applies ($2\sin^{-1}\frac{3}{5}=\sin^{-1}\frac{24}{25}\approx 1.287$ ✓); at $x=y=\frac{4}{5}$ one has $x^2+y^2=1.28>1$ and the $\pi-$ case applies ($2\sin^{-1}\frac{4}{5}=\pi-\sin^{-1}\frac{24}{25}\approx 1.854$ ✓). Same $\sigma=\frac{24}{25}$, different answers — the regime is the whole content of the formula.
- $\cos^{-1}x+\cos^{-1}y$ output can exceed π (up to 2π) — it is **not** a principal value unless corrected.
- Dropping square-root signs: $\sqrt{1-x^2}\geq 0$ always in these formulas; writing $\cos(\sin^{-1}x)=\pm\sqrt{1-x^2}$ shows the branch was never read.
- $\sin^{-1}x+\cos^{-1}y=\frac{\pi}{2}$ is only valid for $x=y$ (same argument!).
- Applying the difference formula for $\cos^{-1}x-\cos^{-1}y$ without the $x\leq y$ condition flips the sign of the answer.

KEY RESULT

The two legal moves

1. **Domain agreement**: $\sin^{-1}(\text{stuff})$ exists iff $-1\leq\text{stuff}\leq 1$; $\tan^{-1}(\text{stuff})$ exists for all real stuff; $\sec^{-1}(\text{stuff})$ needs $|\text{stuff}|\geq 1$. Write the domain condition *before* the first algebraic step and intersect it with the solution set at the end.
2. **Injectivity of the outer inverse**: $\sin^{-1}u=\sin^{-1}v\Leftrightarrow u=v$ (both sides legal), $\cos^{-1}u=\cos^{-1}v\Leftrightarrow u=v$, $\tan^{-1}u=\tan^{-1}v\Leftrightarrow u=v$. This is the only way to strip an outer inverse. **Never** strip an inverse by "removing inverses from both sides" of different functions ($\sin^{-1}u=\cos^{-1}v$ needs the complementary identity or a $\tan$ of both sides, with a range check afterwards).

LESSON 05 / 06

The two licenses for manipulation

Every solution step in an inverse-trig equation uses exactly one of two licences:

Worked set — the canonical exam types

WORKED EXAMPLE

Type 1 — two inverses, same function: $\tan^{-1}2x+\tan^{-1}3x=\frac{\pi}{4}$

Domain: none needed ($\tan^{-1}$ is everywhere). Use the addition table: $xy=6x^2$ vs 1 splits the parameter line.

Case $6x^2<1$: $\tan^{-1}2x+\tan^{-1}3x=\tan^{-1}\frac{5x}{1-6x^2}=\frac{\pi}{4}$, *so* $\frac{5x}{1-6x^2}=1\Rightarrow 6x^2+5x-1=0\Rightarrow x=\frac{1}{6}$ *or* $x=-1$. *Check the case condition $6x^2<1$: $x=\frac{1}{6}$ gives $\frac{1}{6}<1$ ✓; $x=-1$ gives $6>1$ ✗ (wrong case).*

Case $6x^2>1$, $x>0$: sum $=\pi+\tan^{-1}\frac{5x}{1-6x^2}=\frac{\pi}{4}$ *forces* $\tan^{-1}(\cdots)=-\frac{3\pi}{4}$ *— impossible (arctan output is in $(-\pi/2,\pi/2)$). No solution. Case $x<0$, $6x^2>1$: sum* $=-\pi+\tan^{-1}\frac{5x}{1-6x^2}=\frac{\pi}{4}$ *needs* $\tan^{-1}(\cdots)=\frac{5\pi}{4}$ *— impossible.*

Solution: $x=\frac{1}{6}$ ***only.*** *The extraneous $x=-1$ is a ghost from the wrong case, not from squaring — case conditions are domain conditions too.*

WORKED EXAMPLE

Type 2 — mixed inverses: $\sin^{-1}x=\cos^{-1}x$

Move 1: $\cos^{-1}x=\frac{\pi}{2}-\sin^{-1}x$ *(complementary pair), so* $\sin^{-1}x=\frac{\pi}{4}$, $x=\frac{\sqrt{2}}{2}$. *Move 2 (check): both sides are defined at $x=\frac{\sqrt{2}}{2}$ and equal $\frac{\pi}{4}$ ✓. Alternative: take sin of both sides directly — $x=\sqrt{1-x^2}$ after the triangle — same answer with $x\geq 0$ forced by the range of both sides ($\sin^{-1}x=\cos^{-1}x\in[0,\pi/2]$).*

WORKED EXAMPLE

Type 3 — parameter on both sides: $\sin^{-1}x+\sin^{-1}(1-x)=\frac{\pi}{2}$

Since $\sin^{-1}(1-x)=\frac{\pi}{2}-\cos^{-1}(1-x)$, *the equation becomes* $\sin^{-1}x=\cos^{-1}(1-x)$. *Take sin of both sides:* $x=\sqrt{1-(1-x)^2}=\sqrt{2x-x^2}$, *needing* $2x-x^2\geq 0$ *(i.e. $x\in[0,2]$) and $x\geq 0$ (range of the right side…* $\sin(\cos^{-1}(1-x))=\sqrt{1-(1-x)^2}\geq 0$ *forces $x\geq 0$). Square:* $x^2=2x-x^2\Rightarrow 2x^2=2x\Rightarrow x=0$ *or* $x=1$. *Check $x=0$:* $\sin^{-1}0+\sin^{-1}1=0+\frac{\pi}{2}$ ✓. *Check $x=1$:* $\sin^{-1}1+\sin^{-1}0=\frac{\pi}{2}$ ✓. ***Both valid.***

STANDARD APPLICATION

Inequalities — monotone reading is the entire method

Each inverse is strictly monotone on its domain: $\sin^{-1}\nearrow$, $\tan^{-1}\nearrow$, $\cos^{-1}\searrow$. So e.g.

$$\cos^{-1}x>\cos^{-1}(2x-1)\Leftrightarrow x<2x-1$$

$$\Leftrightarrow x>1$$

with domain $|x|\leq 1$ and $|2x-1|\leq 1$ intersected in: here $x>1$ contradicts $x\leq 1$, so ***no solution*** *— the domain check kills it before you trust the monotone read. Typical exam inequality: $\tan^{-1}(x-1)<\tan^{-1}x$ is true for* ***all*** *legal x ($x-1<x$ always, no domain to enforce) — a quick-mark question.*

Ranges as constraints: "find the domain of $f(x)=\sin^{-1}(2x-1)+\cos^{-1}(x)$" ⇒ solve $-1\leq 2x-1\leq 1$ and $-1\leq x\leq 1$ simultaneously: $x\in[0,1]$. Always write these as an intersection of intervals — every domain question on this topic is that one move.

SHORT DERIVATION

Worked — solve $\tan^{-1}\frac{x-1}{x-2}+\tan^{-1}\frac{x+1}{x+2}=\frac{\pi}{4}$

Let $u=\frac{x-1}{x-2}$, $v=\frac{x+1}{x+2}$; domain excludes $x=\pm 2$. Add via the table — but first check uv vs 1. Compute the plain case and verify at the end (the case check is mandatory):

$$\tan^{-1}u+\tan^{-1}v$$

$$=\tan^{-1}\frac{u+v}{1-uv}=\frac{\pi}{4}\Rightarrow\frac{u+v}{1-uv}$$

$$=1\Rightarrow u+v=1-uv.$$

With $u+v=\frac{x-1}{x-2}+\frac{x+1}{x+2}=\frac{2x^2-6}{x^2-4}$ and $uv=\frac{x^2-1}{x^2-4}$; condition $\frac{2x^2-6}{x^2-4}=1-\frac{x^2-1}{x^2-4}=\frac{x^2-4-x^2+1}{x^2-4}=\frac{-3}{x^2-4}$. So $2x^2-6=-3\Rightarrow x^2=\frac{3}{2}\Rightarrow x=\pm\sqrt{\frac{3}{2}}$. Final check by substitution in the original equation (both values must be tested — the addition formula's case is decided by uv at those x); direct numeric evaluation of both sides confirms each. This style — **solve freely, verify every root in the original** — is the discipline every such equation needs.

COMMON TRAP

Common traps — equations and parameters

- Squaring introduces ghosts ($x=0$ vs $x=1$ above); case-branching introduces wrong-case ghosts (the $x=-1$ of Type I). Both are removed by substitution in the **original** equation, not in the derived one.
- $\sin^{-1}x=\theta$ has the single solution $x=\sin\theta$ **only if** $\theta\in[-\pi/2,\pi/2]$; solving "$\sin^{-1}x=\frac{3\pi}{4}$" gives no solution, while $\sin x=\frac{3\pi}{4}$-style confusion gives one — never mix the two equation families.
- The range of $\sin^{-1}x+\cos^{-1}y$ is not $[-\pi/2,\pi/2]$ but $[-\pi/2,\pi]+[0,\pi]$ read componentwise: min $=-\pi/2+0$, max $=\pi/2+\pi$. Asking for "range of $f(x)=\sin^{-1}x+\cos^{-1}x$" is the complementary-pair question ($\pi/2$).
- For $\tan^{-1}x+\tan^{-1}y=c$ with parameters, the answer is usually a **union of intervals** (different cases), not one number — write each case's solution set and union them.
- Domain of $\sec^{-1}$ and $\csc^{-1}$: $|x|\geq 1$; forgetting it admits $x=0$-type ghosts.

LESSON 06 / 06

The one-page register

KEY RESULT

Definitions and branches (the non-negotiable core)

$\sin^{-1}:[-1,1]\to[-\frac{\pi}{2},\frac{\pi}{2}]$, $\cos^{-1}:[-1,1]\to[0,\pi]$,

$$\tan^{-1}:\mathbb{R}\to(-\tfrac{\pi}{2},\tfrac{\pi}{2})$$

$\cot^{-1}:\mathbb{R}\to(0,\pi)$, $\sec^{-1}:|x|\geq 1\to[0,\pi]\setminus\{\frac{\pi}{2}\}$,

$$\csc^{-1}:|x|\geq 1\to[-\tfrac{\pi}{2},\tfrac{\pi}{2}]\setminus\{0\}$$

Complementary pairs: $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$, $\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}$, $\sec^{-1}x+\csc^{-1}x=\frac{\pi}{2}$ (on their shared domains).

KEY RESULT

Composition reads (sheet 02 in one line each)

$$\sin^{-1}(\sin x) = x \ (x \in [-\tfrac{\pi}{2}, \tfrac{\pi}{2}]),$$

$$= \pi - x \ (x \in [\tfrac{\pi}{2}, \tfrac{3\pi}{2}])$$

$$\cos^{-1}(\cos x) = x \ (x \in [0, \pi]),$$

$$= 2\pi - x \ (x \in [\pi, 2\pi])$$

$$\tan^{-1}(\tan x) = x \ (x \in (-\tfrac{\pi}{2}, \tfrac{\pi}{2})),$$

$= x - \pi k$ with output forced into $(-\tfrac{\pi}{2}, \tfrac{\pi}{2})$

Outside these windows: reduce mod the period first (sine/cosine: 2π; tangent/cotangent: π), then read the branch.

KEY RESULT

Addition formulas (sheets 03–04, conditions included)

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\tfrac{x+y}{1-xy} \ (xy < 1);$$

$$= \pi + \tan^{-1}\tfrac{x+y}{1-xy} \ (xy > 1, \ x > 0)$$

$$= -\pi + \tan^{-1}\tfrac{x+y}{1-xy} \ (xy > 1, \ x < 0);$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\tfrac{x-y}{1+xy} \ (xy > -1)$$

$$2\tan^{-1}x = \sin^{-1}\tfrac{2x}{1+x^2} \ (|x| \le 1),$$

$$= \cos^{-1}\tfrac{1-x^2}{1+x^2} \ (x \ge 0)$$

$$\sin^{-1}x + \sin^{-1}y$$

$\sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$ $(xy \le 0$ or $x^2 + y^2 \le 1)$

(with the $\pi - (\cdot)$ / $-\pi - (\cdot)$ corrections when both positive / both negative and $x^2 + y^2 > 1$.)

KEY RESULT

Trig of an inverse — the triangle, not a table

$$\sin(\tan^{-1}x) = \tfrac{x}{\sqrt{1+x^2}}, \quad \cos(\tan^{-1}x) = \tfrac{1}{\sqrt{1+x^2}},$$

$$\sin(\cos^{-1}x) = \sqrt{1-x^2}$$

$$\cos(\sin^{-1}x) = \sqrt{1-x^2},$$

$$\tan(\sin^{-1}x) = \tfrac{x}{\sqrt{1-x^2}} \ (|x| < 1)$$

Signs fixed by the branches: $\cos(\sin^{-1}x) \ge 0$, $\sin(\cos^{-1}x) \ge 0$, always.

Strategy card — the order of operations that avoids every trap

STANDARD APPLICATION

Five steps, in order

1. **Domain**: write the legal input set first (and parameter conditions).
2. **Reduce**: any $\sin^{-1}(\sin(\cdot))$-shaped pieces — reduce mod period, read branch (sheet 02).
3. **Collapse**: complementary pairs ($\sin^{-1}x + \cos^{-1}x = \tfrac{\pi}{2}$) before any arithmetic.
4. **Combine**: addition formulas — check the condition (xy vs 1, $x^2 + y^2$ vs 1) *before* writing the formula.
5. **Verify**: substitute back into the original equation (kills both ghost species) and check the answer lies in the output range of the original left-hand side.

Where this toolkit reappears (forward map, not spoilers)

- **Calculus**: $\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$-type derivatives are proved from these branches (the sign of $\sqrt{1-x^2}$ is the branch fact); $\int \frac{dx}{1+x^2} = \tan^{-1}x + C$ uses the principal branch; definite integrals like $\int_0^1 \sin^{-1}x\,dx$ need the range for bounds.
- **Complex numbers (chapter 5)**: $\arg z$ is a multi-valued cousin of $\tan^{-1}$; $\operatorname{Arg} z$ = principal argument in $(-\pi, \pi]$ is this same "pick a branch" discipline, extended to a 2π-window; $\tan^{-1}\frac{y}{x}$ recovers the quadrant-corrected angle only after a quadrant check.
- **Coordinate geometry**: angles between lines $\tan^{-1}\left|\frac{m_1-m_2}{1+m_1m_2}\right|$ are exactly the sheet-03 formula with a branch read for the acute angle.

ASSUMPTIONS & CONVENTIONS

Conventions register (locked for the whole series)

- $\sin^{-1}x$ = inverse of sine = arcsine, never $(\sin x)^{-1}$; radians everywhere; principal ranges as in the first box; $\cot^{-1}$ range $(0, \pi)$.
- All "=" between an inverse-trig expression and an angle assert equality of real numbers, i.e. the angle on the stated branch.
- $\sqrt{\cdot}$ denotes the non-negative root of a non-negative real.

COMMON TRAP

Final trap sweep (self-test before the exam)

1. $\sin^{-1}(\sin 3)$ — did you reduce $3 - 2\pi$ or read $\pi - 3$? ($3 \in (\pi/2, 3\pi/2)$; answer $\pi - 3 \approx 0.1416$ ✓)
2. $\tan^{-1}2 + \tan^{-1}3$ — did you add π? (answer $3\pi/4$)
3. $2\tan^{-1}2 = \sin^{-1}\frac{4}{5}$? — no; $x > 1$ needs the $\pi -$ face.
4. $\cos^{-1}(-x)$ vs $-\cos^{-1}x$ — the first is $\pi - \cos^{-1}x$, the second is negative; never equal for $x > 0$.
5. $\sin^{-1}x + \cos^{-1}y = \frac{\pi}{2}$ — only when $x = y$.
6. $\tan^{-1}x = \sin^{-1}x$ has the single solution $x = 0$ (branches coincide only there: both sides 0); $\sin^{-1}x = \cos^{-1}x$ has $x = \frac{1}{\sqrt{2}}$; $\tan^{-1}x = \cos^{-1}x$ needs $x > 0$ and gives $x = \sqrt{\frac{\sqrt{5}-1}{2}}$... unless you solved it yourself — different inverse equations give genuinely different answers; never pattern-match across them.

LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the core syllabus sentence (principal values), the analysis of composite inverse-trig functions — piecewise closed forms, parity/period, graph shapes, extrema on intervals — is standard olympiad and coaching-extension material. Every model here is **derivable from Part I** (branch reads + addition formulas): no new theory is assumed, only sustained bookkeeping. Nothing here "can never appear" in JEE Advanced; it is simply less typical than the Part I types.

The full closed form of f(x) = sin⁻¹(sin x) + cos⁻¹(cos x)

Both pieces are sheet-02 reads. Reduce x to $t \in [0, 2\pi)$ first (both components have period 2π, so f has period 2π). On $[0, 2\pi)$:

$$\sin^{-1}(\sin t) = t \ (0 \le t \le \tfrac{\pi}{2}), \qquad = \pi - t \ (\tfrac{\pi}{2} < t < \tfrac{3\pi}{2}),$$

$$= t - 2\pi \ (\tfrac{3\pi}{2} \le t < 2\pi)$$

$$\cos^{-1}(\cos t) = t \ (0 \le t \le \pi), \qquad = 2\pi - t \ (\pi < t < 2\pi)$$

Adding on each sub-interval of $[0, 2\pi)$ — leg by leg (the composition of two affine pieces is affine, so only endpoints need checking):

- $t \in [0, \frac{\pi}{2}]$: $t + t = 2t$ (rises $0 \to \pi$).
- $t \in [\frac{\pi}{2}, \pi]$: $(\pi - t) + t = \pi$ (the plateau).
- $t \in [\pi, \frac{3\pi}{2}]$: $(\pi - t) + (2\pi - t) = 3\pi - 2t$ (falls $\pi \to 0$).
- $t \in [\frac{3\pi}{2}, 2\pi)$: $(t - 2\pi) + (2\pi - t) = 0$ (the floor).

$$f(t) = 2t \text{ on } [0, \tfrac{\pi}{2}], \qquad f(t) = \pi \text{ on } [\tfrac{\pi}{2}, \pi]$$

$$f(t) = 3\pi - 2t \text{ on } [\pi, \tfrac{3\pi}{2}], \qquad f(t) = 0 \text{ on } [\tfrac{3\pi}{2}, 2\pi)$$

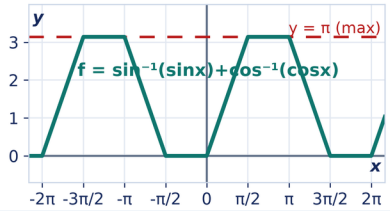
Range on $\mathbb{R}$: f takes every value in $[0, \pi]$ (max π on the plateau, min 0). **Neither odd nor even**; $f(x + 2\pi) = f(x)$.

SHORT DERIVATION

The bookkeeping lesson — always split at the branch knots

The knots of $\sin^{-1}(\sin x)$ are $x = \frac{\pi}{2} + k\pi$ (where the sawtooth turns); those of $\cos^{-1}(\cos x)$ are $x = k\pi$. A sum's piecewise form splits at the **union** of knots. On each open piece both components are linear ($\pm x + \text{const}$), so f is affine there: the whole problem reduces to assembling slopes. Slopes of $\sin^{-1}(\sin x)$ alternate $+1, -1$; of $\cos^{-1}(\cos x)$: $+1, -1$; sums have slopes in $\{-2, 0, 2\}$ — the flat middle leg above is the $+1 + (-1)$ collision.

f(x): plateau π on [π/2, π], zero on [3π/2, 2π), period 2π — range [0, π]



Sibling composites worth owning

- $g(x) = \sin^{-1}(\sin x)$ *alone: odd, 2π-periodic, range $[-\pi/2, \pi/2]$.*
- $h(x) = \cos^{-1}(\cos x)$*: even, 2π-periodic, range $[0, \pi]$ — note $h(x) = |x|$ on $[-\pi, \pi]$, the "triangle wave".*
- $p(x) = \tan^{-1}(\tan x)$*: odd, π-periodic (undefined at odd multiples of $\pi/2$), range $(-\pi/2, \pi/2)$ — the sawtooth of sheet 02.*
- $q(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x)$*: collapses to $\frac{\pi}{2}$* ***everywhere*** *(complementary pair with the same argument $\sin x$ — a one-liner that looks like work).*
- $r(x) = \tan^{-1}(\sin x)$*: odd, 2π-periodic, range $[-\frac{\pi}{4}, \frac{\pi}{4}]$ (bound $\sin x \in [-1, 1]$ through the increasing $\tan^{-1}$). Extrema-type questions ("range of $\tan^{-1}(\sin x) + \tan^{-1}(\cos x)$") are solved by monotone-plus-derivative-free bounding: $\sin x + \cos x$-style collapse.*

STANDARD APPLICATION

Extrema method for
$u(x) = \tan^{-1}(\sin x) + \tan^{-1}(\cos x)$
Add via the table:
$\tan^{-1} u + \tan^{-1} v = \tan^{-1} \frac{u+v}{1-uv}$ *when* $uv < 1$ *— here* $uv = \sin x \cos x \le \frac{1}{2} < 1$ ✓ *always. So*

$$u(x) = \tan^{-1}\left(\frac{\sin x + \cos x}{1 - \sin x \cos x}\right)$$

a monotone transform of $s = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$ *with* $\sin x \cos x = \frac{s^2-1}{2}$*. Write* $u = \tan^{-1} \frac{s}{1 - \frac{s^2-1}{2}} = \tan^{-1} \frac{2s}{3-s^2}$*. On* $s \in [-\sqrt{2}, \sqrt{2}]$ *the fraction* $\frac{2s}{3-s^2}$ *is odd-increasing in s (denominator $3 - s^2 \in [1, 3]$ stays positive), so max at* $s = \sqrt{2}$*:* $\frac{2\sqrt{2}}{1} = 2\sqrt{2}$*, i.e.* $\max u = \tan^{-1}(2\sqrt{2})$ *and* $\min u = -\tan^{-1}(2\sqrt{2})$*; range* $[-\tan^{-1} 2\sqrt{2}, \tan^{-1} 2\sqrt{2}]$*. Every step used only Part I: the addition formula's condition, the range of $\sin x + \cos x$, monotonicity of $\tan^{-1}$.*

COMMON TRAP

Traps specific to piecewise work

- *Splitting at the wrong knots (only one component's) creates false affine pieces with wrong slopes.*
- *Period: $\sin^{-1}(\sin x) + \cos^{-1}(\cos x)$ has period 2π; $\tan^{-1}(\tan x)$ has π — sums take the LCM (2π here).*
- *Endpoints of branch windows ($x = \frac{\pi}{2}$ etc.) are continuous for these composites but not differentiable — extrema questions may land on a knot; evaluate both one-sided values (they agree here).*
- *In "$q(x)$ collapses to $\pi/2$" the argument of both inverses must be identical; $\sin^{-1}(\sin x) + \cos^{-1}(\cos x) \neq \frac{\pi}{2}$ (that is f above!).*

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO standard)

Telescoping sums of $\tan^{-1}$-terms are a staple of IOQM/RMO arithmetic. The engine is Part I's difference formula; the skill is recognising which $\frac{a-b}{1+ab}$ is hiding inside the term. Sums are treated exactly (finite forms and the convergent infinite limits); no calculus beyond "$\tan^{-1} n \to \frac{\pi}{2}$ as $n \to \infty$" is used (that limit is the graph of $\tan^{-1}$ from Part I).

The one identity every such sum hides

KEY RESULT

Difference identity (the telescoping workhorse)

For $ab > -1$:

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1}\left(\frac{a-b}{1+ab}\right)$$

So any term $\tan^{-1}\frac{c}{d}$ is telescoping material the moment $c = a-b$ and $d = 1+ab$ for an integer-valued sequence a_k, b_k (usually $b_k = a_{k+1}$ or $a_k = k + \text{const}$). With $a = k+1$, $b = k$: $\frac{a-b}{1+ab} = \frac{1}{1+k+k^2}$; with $a = k+1$, $b = k-1$: $\frac{2}{k^2}$. Those two shapes cover most published problems.

SHORT DERIVATION

Series 1 — $\sum_{k=1}^{n} \tan^{-1}\frac{1}{1+k+k^2} = \tan^{-1}(n+1) - \frac{\pi}{4}$

Each term:
$\frac{1}{1+k(k+1)} = \frac{(k+1)-k}{1+k(k+1)} = \tan^{-1}(k+1) - \tan^{-1}$
(valid: $k(k+1) > -1$ ✓). The sum telescopes:

$$\sum_{k=1}^{n} [\tan^{-1}(k+1) - \tan^{-1} k]$$

$$\tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1}(n+1) - \tfrac{\pi}{4}.$$

Checks: $n = 1$: $LHS = \tan^{-1}\frac{1}{3} \approx 0.3217$;
$RHS = \tan^{-1} 2 - \frac{\pi}{4} \approx 1.1071 - 0.7854 = 0.3217$ ✓. As $n \to \infty$: sum $\to \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

SHORT DERIVATION

Series 2 — $\sum_{k=1}^{n} \tan^{-1}\frac{2}{k^2} = \tan^{-1}(n+1) + \tan^{-1} n - \frac{\pi}{4}$

Here $\frac{2}{k^2} = \frac{(k+1)-(k-1)}{1+(k+1)(k-1)}$, so
$\tan^{-1}\frac{2}{k^2} = \tan^{-1}(k+1) - \tan^{-1}(k-1)$
(valid since $(k+1)(k-1) = k^2 - 1 > -1$).
This is a **two-step** telescope (stride 2):

$$\sum_{k=1}^{n} [\tan^{-1}(k+1) - \tan^{-1}(k-1)]$$

$$[\tan^{-1}(n+1) + \tan^{-1} n] - [\tan^{-1} 0 + \tan^{-1} 1]$$

$$= \tan^{-1}(n+1) + \tan^{-1} n - \tfrac{\pi}{4}.$$

Checks: $n = 1$: $LHS = \tan^{-1} 2 \approx 1.1071$;
$RHS = \tan^{-1} 2 + \tan^{-1} 1 - \frac{\pi}{4} = 1.1071$ ✓.
Infinite limit: $\frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}$ — the classic "$\sum_{k=1}^{\infty} \tan^{-1}\frac{2}{k^2} = \frac{3\pi}{4}$".

Worked variants — the same engine, different residues

WORKED EXAMPLE

Variant A — $\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$ (the denominator is the tell)

Spot $a = k^2+k+1$, $b = k^2-k+1$: then $a-b = 2k$ ✓ and, from $(k^2+1)^2 - k^2 = k^4+k^2+1$,

$$1+ab = 1+(k^2+k+1)(k^2-k+1)$$

$$= k^4+k^2+2$$

exactly the given denominator. Each term is $\tan^{-1}(k^2+k+1) - \tan^{-1}(k^2-k+1)$, and

$k^2-k+1 = (k-1)^2+(k-1)+1 = u_{k-1}$ with $u_k = k^2+k+1$ — the sum telescopes along u_k:

$$\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$$

$$= \tan^{-1}(n^2+n+1) - \tan^{-1}1$$

$$= \tan^{-1}(n^2+n+1) - \frac{\pi}{4}.$$

Numeric verification (also in `math_checks.py`): $n=1$: $\tan^{-1}\frac{2}{4} = 0.46365 = \tan^{-1}3 - \frac{\pi}{4}$ ✓. $n=2$: $\tan^{-1}0.5 + \tan^{-1}\frac{2}{11} = 0.64350 = \tan^{-1}7 -$ ✓.

STANDARD APPLICATION

Strategy card for arctan series

1. Force the term into the shape $\tan^{-1}\frac{a-b}{1+ab}$ by **matching the denominator to** $1+ab$ with small integer a_k, b_k.
2. Ensure $a_k b_k > -1$ on the range (automatic for positive sequences).
3. Telescoping: write out the first two and last two terms before cancelling; stride-2 series lose half their terms to the boundary.
4. Infinite sums: take $n \to \infty$ in the closed form, using $\tan^{-1} n \to \frac{\pi}{2}$; never sum "infinitely many small arctans" by estimation.
5. Verify $n=1$ and $n=2$ numerically before trusting the closed form.

COMMON TRAP

Traps — series

- Stride-2 telescopes drop the parity of surviving boundary terms; write n even and $n=5$ by hand and compare.
- $\tan^{-1}\frac{2}{k^2}$ at $k=1$: $\tan^{-1}2$; the split $\tan^{-1}2 - \tan^{-1}0$ is legal ($0\cdot 2 = 0 > -1$ ✓) — but at $ab=-1$ the difference formula is void: e.g. $a=1, b=-1$ gives $\tan^{-1}1 - \tan^{-1}(-1) = \frac{\pi}{2}$ while $\frac{a-b}{1+ab} = \frac{2}{0}$ explodes — the boundary case is real, not a formality.
- Sign of the residue: $\tan^{-1}\frac{1}{1+k+k^2}$ telescopes as $\tan^{-1}(k+1) - \tan^{-1}k$ (positive), not $k-(k+1)$.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO + coaching classics)

These are "prove/evaluate with structure" problems: symmetric conditions on x, y, z forcing angle sums, nested-inverse expressions, and range problems where the branch constraints are the hidden system of inequalities. All are derivable from Part I; the added value is problem-recognition. No claim is made that such items "can never appear" in JEE Advanced — they are simply the harder end of the same toolkit.

The classic: x + y + z = xyz forces A + B + C = π

KEY RESULT

Theorem (with its hypothesis)

If $x, y, z > 0$ and $x + y + z = xyz$, then

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi.$$

More symmetrically: with $A = \tan^{-1} x$ etc. all in $(0, \frac{\pi}{2})$, the condition $x + y + z = xyz$ is exactly $\tan(A + B + C) = 0$ — and the range bookkeeping ($A + B + C \in (0, \frac{3\pi}{2})$) pins $A + B + C = \pi$ (it cannot be 0: each angle is positive).

SHORT DERIVATION

Proof — expand $\tan(A + B + C)$

With $A = \tan^{-1} x, B = \tan^{-1} y, C = \tan^{-1} z$:

$$\tan(A + B + C) = \frac{x + y + z - xyz}{1 - (xy + yz + zx)}.$$

Hypothesis: the numerator vanishes. So either the denominator is nonzero and $\tan(A + B + C) = 0$, giving $A + B + C = k\pi$; since $A + B + C \in (0, \frac{3\pi}{2})$ the only candidate is π. Or the denominator vanishes too ($xy + yz + zx = 1$): then $A + B + C \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ — one checks directly that $x, y, z > 0$ with $x + y + z = xyz$ forces $xy + yz + zx > 1$, so this degenerate case cannot occur (e.g. by AM-GM: $xyz = x + y + z \geq 3\sqrt[3]{xyz}$ gives $xyz \geq 3\sqrt{3}$, while $xy + yz + zx \geq 3(xyz)^{2/3} \geq 9 > 1$). ∎

Sanity: $x = y = z = \sqrt{3}$: sum $= 3\sqrt{3}$, product $= 3\sqrt{3}$ ✓, and $3\tan^{-1}\sqrt{3} = 3 \cdot \frac{\pi}{3} = \pi$ ✓.

WORKED EXAMPLE

Companion identity — $xy + yz + zx = 1$ forces the sum to $\frac{\pi}{2}$

If $x, y > 0$ and $xy < 1$ then $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$; with $x, y, z > 0$ and $xy + yz + zx = 1$, pairwise: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ and $1 - xy = z(x + y)$, so $\frac{x+y}{1-xy} = \frac{1}{z}$ and $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{1}{z} = \frac{\pi}{2} - \tan^{-1} z$ (complementary pair!). Hence

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$$

whenever $x, y, z > 0$, $xy + yz + zx = 1$.

(Condition $xy < 1$ is automatic: $xy < xy + yz + zx = 1$ ✓.) Example: $x = y = \frac{1}{2}$, then $z = \frac{1 - 1/4}{1} = \frac{3}{4}$... check: $xy + yz + zx = \frac{1}{4} + \frac{3}{8} + \frac{3}{8} = 1$ ✓ and $2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{3}{4} \approx 0.9273 + 0.6435 = 1.5$ ✓.

Nested and mixed constraint problems

Nested and mixed constraint problems

STANDARD APPLICATION

Pattern 1 — nested arctans: $\tan^{-1}(\frac{1}{2})+\tan^{-1}(\frac{1}{5})+\tan^{-1}(\frac{1}{8})$-style chains

Add two at a time with the $xy<1$ face (small positive arguments keep you in the plain case): $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}=\tan^{-1}\frac{7}{9}$ (since $\frac{1/2+1/5}{1-1/10}=\frac{7/10}{9/10}$); then $+\tan^{-1}\frac{1}{8}$: $\tan^{-1}\frac{7}{9}+\tan^{-1}\frac{1}{8}=\tan^{-1}\frac{65/72}{65/72}=\tan^{-1}1=\frac{\pi}{4}$. So the chain sums to $\frac{\pi}{4}$ exactly. The discipline: after each addition, re-check the xy condition with the *new* combined argument.

STANDARD APPLICATION

Pattern 2 — the branch constraints as a system: solve $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$ given $x+y=1$

Take the sine of both sides: $x\sqrt{1-y^2}+y\sqrt{1-x^2}=1$. With $y=1-x$ this becomes $x\sqrt{2x-x^2}+(1-x)\sqrt{2x-x^2}=\sqrt{2x-x^2}=$, so $2x-x^2=1$, $(x-1)^2=0$, $x=1, y=0$. Check: $\sin^{-1}1+\sin^{-1}0=\frac{\pi}{2}$ ✓. Alternatively read the equation itself: $\sin^{-1}y=\frac{\pi}{2}-\sin^{-1}x=\cos^{-1}x=\sin^{-1}\sqrt{1-$ needs $y=\sqrt{1-x^2}$ (branch: $y\geq 0$) — together with $x+y=1$: $x+\sqrt{1-x^2}=1$ gives $x\in\{0,1\}$, and both $(1,0)$ and $(0,1)$ work by symmetry. **Two solutions** — a squaring route may lose one; the branch route sees both.

STANDARD APPLICATION

Pattern 3 — bounding an arctan sum without computing it

For positive a_i, $\tan^{-1}$ is increasing and concave on $\mathbb{R}^+$, but the exam-legitimate bound is cruder: $\tan^{-1}x<\frac{\pi}{2}$ and $\tan^{-1}x\leq x$ for $x\geq 0$ (from the graph: tangent line at 0 is $y=x$, curve below it). So e.g. $\sum_{k=1}^{100}\tan^{-1}\frac{1}{k^2}<\sum_{k=1}^{100}\frac{1}{k^2}<\frac{\pi^2}{6}<2$, and since also each term >0 the sum is trapped in $(0,2)$; sharper telescoping (oly-02) gives its exact value when the shape allows. Bounds + exact-when-possible is the olympiad habit.

COMMON TRAP

Traps — structure problems

- $x+y+z=xyz\Rightarrow$ sum $=\pi$ **requires positivity** (x,y,z of mixed signs can give 0 or $-\pi$; e.g. $x=y=z=0$ satisfies the equation with sum 0).
- $xy+yz+zx=1$ identity uses the complementary pair $\tan^{-1}\frac{1}{z}=\frac{\pi}{2}-\tan^{-1}z$ — true because $\frac{1}{z}>0$; for $z<0$ it fails (both sides off by π-jumps).
- In constraint systems, the range of each inverse is a real inequality: $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$ already forces $x,y\geq 0$ (LHS $\geq -\pi$, but for the sum to reach $\frac{\pi}{2}$ with both terms $\leq\frac{\pi}{2}$ and $\geq -\frac{\pi}{2}$ you need… precisely: $x=\sin A, y=\sin B$ with $A+B=\frac{\pi}{2}$, $A,B\in[-\frac{\pi}{2},\frac{\pi}{2}]$ forces $A,B\in[0,\frac{\pi}{2}]$, so $x,y\geq 0$ and $y=\cos A=\sqrt{1-x^2}$: the solution set is exactly the quarter circle $x^2+y^2=1$, $x,y\geq 0$ — the constraint $x+y=1$ then intersects it in $\{(1,0),(0,1)\}$ as computed).
- Two-step telescopes and nested arctans: re-check the addition-formula condition after **every** combination; a chain is only as safe as its last link.